**Log File Analyzer for Intrusion Detection**

**Introduction**

In today's digital environment, monitoring system activity is crucial to ensure security. Log files contain important records of events such as login attempts and user actions. Analyzing these logs helps in identifying suspicious behavior and potential security threats. This project focuses on building a simple log file analyzer using Python to detect failed login attempts and identify suspicious IP addresses.

**Abstract**

The Log File Analyzer is a Python-based tool designed to process and analyze log files for intrusion detection. It reads log data from a text file, extracts IP addresses using regular expressions, and identifies failed login attempts. The system counts repeated login failures and highlights suspicious IP addresses that may indicate brute-force attacks. The results are displayed on the console and saved in a report file for further analysis.

**Tools Used**

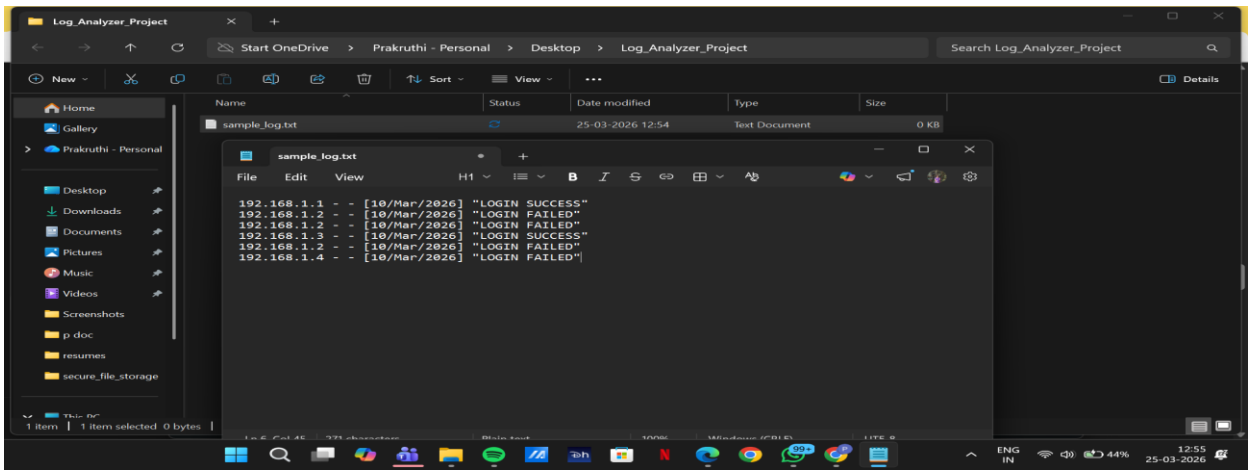
- Python
- Regular Expressions (re module)
- Collections (Counter)
- Text File Handling

**Steps Involved in Building the Project**

1. Created a sample log file containing login activity data.
2. Wrote a Python script to read and process the log file.
3. Used regular expressions to extract IP addresses from each log entry.
4. Identified and counted failed login attempts.
5. Used Counter to find repeated IP addresses.
6. Displayed suspicious IPs with multiple failed attempts.
7. Generated and saved the analysis report in a text file.

**Conclusion**

The project successfully demonstrates how log analysis can be used for basic intrusion detection. By identifying repeated failed login attempts and suspicious IP addresses, the system helps in detecting potential security threats. This simple tool can be further enhanced with advanced features like real-time monitoring, GUI, and integration with security systems.



main.py - C:/Users/prakr/OneDrive/Desktop/Log_Analyzer_Project/main.py (3.11.4)

File Edit Format Run Options Window Help

```
    logs = file.readlines()

failed_attempts = 0
ip_list = []

# Process logs
for line in logs:
    # Extract IP address
    ip = re.findall(r'\d+\.\d+\.\d+\.\d+', line)
    if ip:
        ip_list.append(ip[0])

    # Check failed login
    if "FAILED" in line:
        failed_attempts += 1

# Count repeated IPs
ip_count = Counter(ip_list)
```

IDLE Shell 3.11.4

File Edit Shell Debug Options Window Help

```
Python 3.11.4 (tags/v3.11.4:d2340ef, Jun  7 2023, 05:45:37) [MSC v.1934 64 bit (AMD64)] o
n win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
== RESTART: C:/Users/prakr/OneDrive/Desktop/Log_Analyzer_Project/main.py ==

🔍 Log Analysis Report
----------------------
Total Failed Logins: 4

⚠️ Suspicious IPs (more than 2 attempts):
192.168.1.2 -> 3 attempts

✅ Report saved as report.txt
>>>
```

Log_Analyzer_Project

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Search Log_Analyzer_Project

New Sort View Details

| Name | Status | Date modified | Type | Size |
|---|---|---|---|---|
| main.py | | 25-03-2026 12:57 | Python File | 2 KB |
| report.txt | | 25-03-2026 12:57 | Text Document | 1 KB |
| sample_log.txt | | 25-03-2026 12:56 | Text Document | 1 KB |

report.txt

File Edit View

```
Log Analysis Report
Failed Logins: 4
Suspicious IPs:
192.168.1.2 -> 3 attempts
```

3 items | 1 item selected 83 bytes | Sync pendi